

Quadrotor Flyability & Endurance Report

PX4 SITL + Gazebo Harmonic + ROS 2 Humble

Version	v1.0.0
Date	2026-09-30
Repository	github.com/shaashwatmishra203-creator/Drone_sim
Airframe	n360_quad - quad X, 495 mm diagonal
Estimated AUW	3.117 kg (88% still unweighed)
Verdict	FLIES. Endurance is the constraint, not thrust.

What this report is, and is not

These results describe a **model** of the aircraft, validated in software-in-the-loop simulation. They do **not** establish that the real aircraft is safe to build, power or fly. No propeller has been spun, no pack discharged, and no thrust measured on a stand. Critically, **88% of the mass budget is still estimated rather than weighed**, and endurance moves roughly 2 minutes per 500 g. Treat every figure as a band, not a verdict.

Prepared with Claude Opus 5. All figures generated directly from config/airframe.yaml, out/sizing.json and the recorded flight logs.

1. Executive summary

The question was whether the parts list the team settled on can fly, and for how long. It can. Thrust-to-weight is about **3.45** and hovering consumes only **29%** of available thrust, which is comfortable margin. Flight capability is not the problem.

Endurance is. In still air the aircraft held a 5 m hover for **11.39 minutes** before reaching the 80% depth-of-discharge limit. In 6 m/s wind with gusts that falls to roughly 9 minutes. For an autonomy platform carrying two stereo cameras and a gimbal, that is a short working window.

Measured endurance	Hover power	Thrust-to-weight	Wind penalty
11.4 min	571 W	3.45	+22% power

Three findings block the build as specified

None of them is about flight dynamics:

#	Finding	Consequence
F1	The ZED 2 and ZED Mini cannot run on a Raspberry Pi 5. The ZED SDK requires an NVIDIA CUDA GPU; the Pi 5 has none.	Perception stack is non-functional as specified.
F2	The SpeedyBee F405 V5 cannot run a PX4 autonomy stack. PX4 dropped F405 support.	Nothing validated here transfers to that board.
F3	The 9x4.5 propellers reach ~15,900 rpm at full throttle against an estimated 16,111 rpm ceiling - about 1% margin.	13-inch props would add 57% endurance.

Evidence quality on F3

The 16,111 rpm ceiling is the **145000/D rule of thumb** for composite propellers, not a figure published by HQProp. This finding is only as strong as that number. Confirm the real rating with the manufacturer before treating it as settled - if their limit is higher, F3 shrinks from a blocking concern to a footnote.

2. The ROS 2 software

The simulation is driven by a small ROS 2 Humble package, `drone_eval`, living in `~/ws_px4/src/` alongside the standard `px4_msgs` and `px4_ros_com`. Data reaches ROS through PX4's uXRCE-DDS bridge rather than a `ros_gz` bridge, which is deliberately not installed - the PX4 path already carries everything needed and is lighter.

```
Gazebo Harmonic <---> PX4 SITL <---> MicroXRCEAgent <---> ROS 2
      (physics)         (flight control)   (DDS bridge)      drone_eval nodes
```

The two nodes

Node	What it does
flight_logger	Subscribes to position, attitude, status, per-motor commands and battery. Converts each motor command to rotor speed, then to thrust and shaft power through the propeller coefficients, and integrates a physical battery model. Writes one CSV per run and latches <code>/drone_eval/battery_empty</code> when the depth-of-discharge gate is reached.
mission_runner	Streams offboard heartbeats, arms, takes off, then flies the scenario profile - hover, forward-speed steps, maximum climb, or waypoints - and lands on completion, battery-empty, or timeout. PX4 requires a stream of setpoints <i>before</i> it will accept the offboard mode switch, so the node warms up before arming.

Why the logger computes power itself

PX4's SITL battery simulator drains the pack linearly on **wall clock** via `SIM_BAT_DRAIN` and never looks at the motors, so its `BatteryStatus` is useless for endurance. Instead the logger takes the real motor commands and integrates an open-circuit-voltage curve with internal-resistance sag. That matters because both effects shorten a flight and a naive energy-divided-by-power calculation misses them: as the pack empties its voltage falls, so the same power demands more current, which sags the voltage further.

Topics and QoS

Topic	Message	Direction
<code>/fmu/out/vehicle_local_position_v1</code>	<code>VehicleLocalPosition</code>	subscribe
<code>/fmu/out/vehicle_attitude</code>	<code>VehicleAttitude</code>	subscribe
<code>/fmu/out/vehicle_status_v4</code>	<code>VehicleStatus</code>	subscribe
<code>/fmu/out/actuator_motors</code>	<code>ActuatorMotors</code>	subscribe
<code>/fmu/out/battery_status_v1</code>	<code>BatteryStatus</code>	subscribe
<code>/fmu/in/offboard_control_mode</code>	<code>OffboardControlMode</code>	publish
<code>/fmu/in/trajectory_setpoint</code>	<code>TrajectorySetpoint</code>	publish
<code>/fmu/in/vehicle_command</code>	<code>VehicleCommand</code>	publish
<code>/drone_eval/battery_empty</code>	<code>std_msgs/Bool</code>	internal, latched

Two traps that cost real debugging time

QoS. PX4 offers every `/fmu/out/` topic as `BEST_EFFORT` with `TRANSIENT_LOCAL` durability. The rclpy default is `RELIABLE`, which is incompatible - a default subscriber receives nothing, silently. An empty log is a QoS mismatch until proven otherwise.

Topic names. PX4 1.18 appends message-version suffixes to some topics and not others, so both nodes resolve names against the live graph instead of hardcoding them. There is also no `/c lock` without a `ros_gz` bridge, so `use_sim_time` would freeze both nodes at zero; they use PX4 message timestamps, which are the simulation clock.

3. Parts and mass register

Every component, with the mass actually used by the model and a column for the real weight once the team measures it. **The measured column is deliberately empty: nothing has been weighed yet.** This is the single largest source of uncertainty in the entire report.

Component	Qty	Unit (g)	Est. (g)	Measured (g)	Confidence	Power (W)
Ovonic 22.2V 6S 50C 6000mAh (XT90 female)	1	830	830	—	estimated	-
HobbyWing XRotor 3115 900KV	4	195	780	—	estimated	-
Frame, PETG-CF printed, + nylon legs	1	550	550	—	estimated	-
3-DoF gimbal + gimbal camera	1	210	210	—	estimated	5.0
Stereolabs ZED 2	1	180	180	—	vendor spec	2.0
ALFA AWUS036ACM dual-band USB adapter + antennas	1	170	170	—	estimated	3.0
Wiring, XT90, standoffs, fasteners, straps	1	120	120	—	estimated	-
Raspberry Pi 5 (8 GB) + active cooler + 5V regulator	1	100	100	—	vendor spec	12.0
Stereolabs ZED Mini	1	62	62	—	vendor spec	2.0
HQProp 9x4.5 nylon composite	4	10	40	—	estimated	-
GPS / compass module + mast	1	40	40	—	estimated	0.5
SpeedyBee F405 V5 stack (FC + 4-in-1 ESC, 30x30)	1	35	35	—	vendor spec	2.0
TOTAL (all-up weight)			3117	—		26.5

Estimated versus real mass

2740 g of 3117 g (88% of all-up weight) is an engineering estimate, not a measurement. The four worth weighing first are the battery (830 g assumed), the four motors (195 g each assumed), the printed frame and legs (550 g assumed) and the gimbal assembly (210 g assumed). Endurance moves about 2 minutes per 500 g, so the true figure could plausibly sit anywhere between roughly 8 and 14 minutes. Filling in the measured column and re-running tools/sizing.py collapses that range to a single number.

Mass sensitivity

What the uncertainty above actually costs, for the current 9-inch build:

All-up weight (kg)	Hover power (W)	Endurance (min)	T/W
2.40	414	15.7	4.49
2.80	516	12.6	3.85
3.20	624	10.3	3.37
3.60	739	8.6	2.99
4.00	861	7.4	2.69

4. Propulsion, battery and mass properties

Motor	HobbyWing XRotor 3115 900KV x4 - 900 KV, 40 A continuous (all values unconfirmed)
Propeller	HQProp 9x4.5 nylon composite x4 - Ct 0.112, Cp 0.048, figure of merit 0.58
ESC	SpeedyBee F405 V5 4-in-1 ESC - 45 A continuous
Battery	Ovonic 22.2V 6S1P 50C 6000mAh - 6S, 6.0 Ah, 136 Wh at 22.7 V nominal
Discharge limit	80% depth of discharge (109 Wh usable)
Disc loading	19.0 kg per square metre (0.164 m2 total disc area)
Inertia (computed)	Ixx 0.0342, Iyy 0.0380, Izz 0.0582 kg m2
Centre of gravity	x +2.6 mm, y -1.6 mm, z +18.7 mm

Propeller comparison

Propeller	Disc load	Hover W	Endur. min	Gain	Max rpm	rpm margin	T/W
9x4.5 (current)	19.0	601	10.7	-	15927	1.1%	3.45
11x4.5	12.8	469	13.8	+29%	10310	21.8%	3.01
13x4.4	9.2	388	16.8	+57%	7334	34.2%	2.76

Disc load in kg per square metre. All configurations pass the thrust gates - the 9-inch rpm margin is the one marginal figure.

5. Flight test results

Two models were built deliberately independently - closed-form momentum theory, and PX4 flying the aircraft in Gazebo with power derived from real motor commands. They share only the configuration file. Their agreement is the reason the rest of these numbers are worth quoting.

Validation: analytical versus simulation

Quantity	Analytical model	Measured in SITL	Difference
Rotor speed (hover)	8569 rpm	8619 rpm	0.6%
Thrust per motor	779 g	788 g	1.2%
Hover power	601 W	571 W	5.0%
Endurance	10.7 min	11.39 min	6.3%

The residual gap is expected: the analytical side uses momentum theory with a figure of merit, while the logger integrates the power coefficient against actual rotor speed. They are different approximations, so exact agreement would be suspicious.

How long it flew

The aircraft held a 5 metre hover in still air until the battery model reached the 80% depth-of-discharge gate, consuming **107.5 Wh** of the 136 Wh pack:

Condition	Endurance	Power (W)	Current (A)	Rotor (rpm)	Thrust/motor (g)	Hold RMS (m)
Still air (default world)	11.39 min	571	25.5	8619	788	0.02
6 m/s wind + 3 m/s gusts	~9.2 min (proj.)	699	30.0	9237	907	0.93
Wind penalty	-19%	+22%	+18%	+7%	+15%	48x worse

Do not plan missions on the still-air number

The 11.4 minute figure is a benchmark measured with zero disturbance. In moderate wind the same aircraft loses about a fifth of its endurance and its position-hold error rises from 2 cm to nearly a metre. **Plan real missions around 7 to 8 minutes of usable time with reserve.**

Scenarios not yet run

This version covers still-air endurance and wind. Four scenarios are built and scripted but have not been executed: full 13-inch endurance, the Jetson compute variant, a forward-speed sweep, a maximum-climb test, and a forest waypoint mission. An interrupted 13-inch run did produce one useful reading - 384.8 W hover against the model's 388 W predicted, a 0.8% match that independently exercises the motor-constant derivation at a different propeller diameter. Re-run everything with `bash scripts/batch2.sh`.

6. Recommended actions

Priority	Action	Cost	Benefit
1	Weigh the aircraft and fill in the measured column in section 3, then re-run the sizing tool.	free	Removes the largest uncertainty in every number here
2	Confirm HQProp's real maximum rpm for the 9x4.5.	free	Settles or dismisses finding F3 entirely
3	Move to 13-inch propellers. The 495 mm frame fits them with about 20 mm to spare - verify against CAD.	one prop set	+57% endurance, 10.7 to 16.8 minutes
4	Swap the Raspberry Pi 5 for a CUDA-capable board such as a Jetson Orin NX.	board	Makes the perception stack possible at all; costs ~0.6 min
5	Swap the SpeedyBee F405 for an H7 flight controller.	board	Makes a PX4 autonomy stack possible at all
6	Specify a dedicated 5 V / 5 A BEC and a power module.	small	Avoids brownout; enables real battery telemetry
7	Thrust-stand the motor and propeller to measure real Ct and Cp.	one day	Collapses the biggest remaining source of model error

What would change these conclusions

Measured masses materially different from the estimates, most likely the printed frame which could plausibly be 400 g or 800 g. An HQProp rating above roughly 17,000 rpm would retire F3. Measured propeller coefficients could move hover power by as much as 20%, carrying every endurance figure with them. None of these would threaten flyability itself - the thrust margin is large enough to absorb them.

A note on what the validation proves

The close agreement between the analytical model and the simulation demonstrates **internal consistency, not physical accuracy**. Both models read the same configuration file, so if the propeller coefficients are wrong then both are wrong together and would still agree with each other. Only a thrust stand can settle that. This is why item 7 above matters more than its low priority suggests.